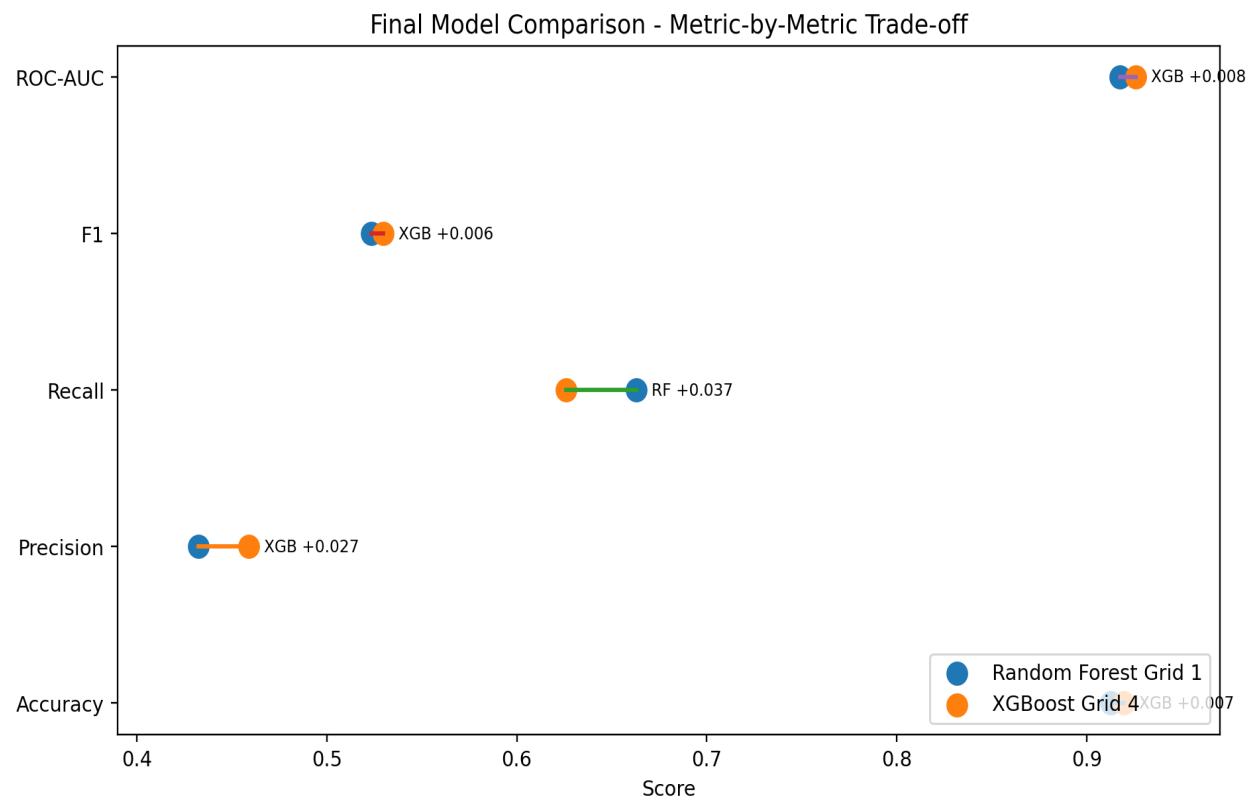


Bank Term Deposit - Comparative Figures

Creative decision-focused figures with interpretation and standardized notebook references

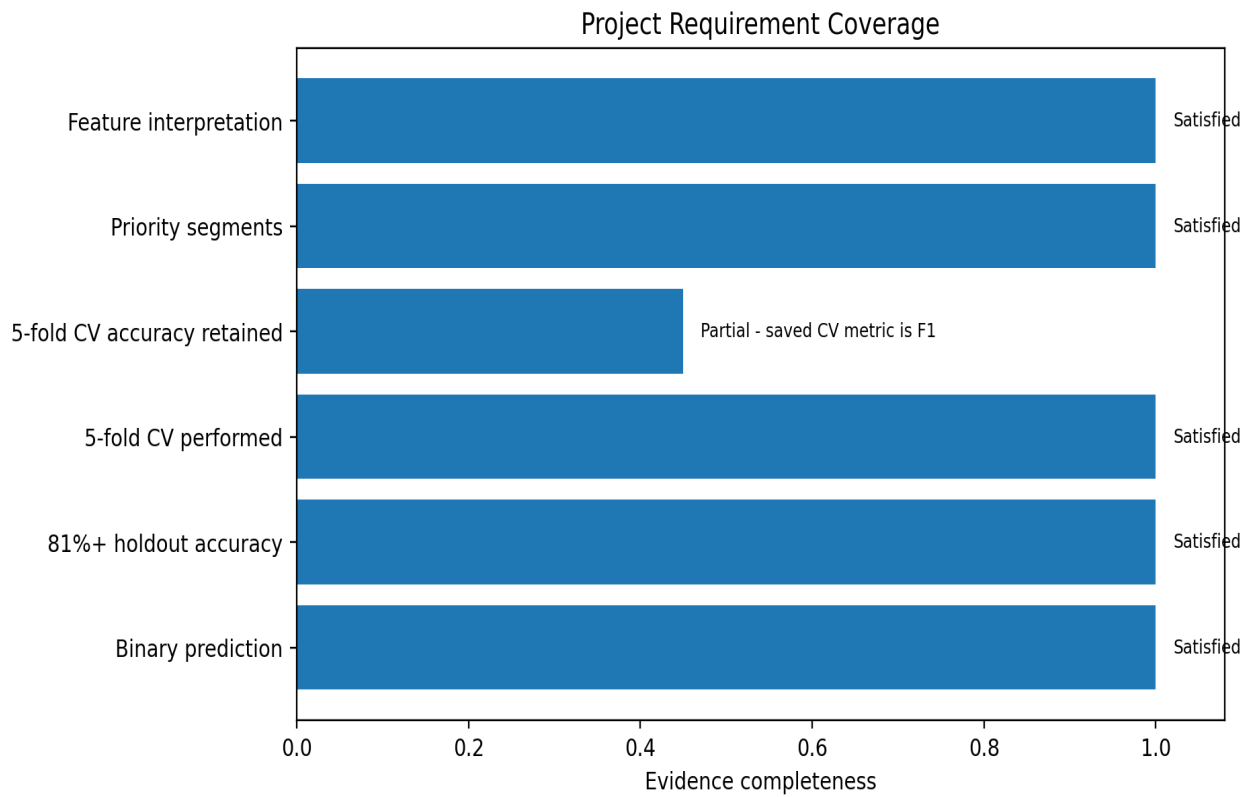
1. Final Model Trade-off



Why this figure matters: Shows the exact metric-by-metric difference between the two final candidates. XGBoost wins most metrics; Random Forest wins Recall.

Source: 02_Random_Forest_Modeling.ipynb and 03_XGBoost_Modeling.ipynb; consolidated in 04_Model_Comparison_and_Business_Analysis.ipynb.

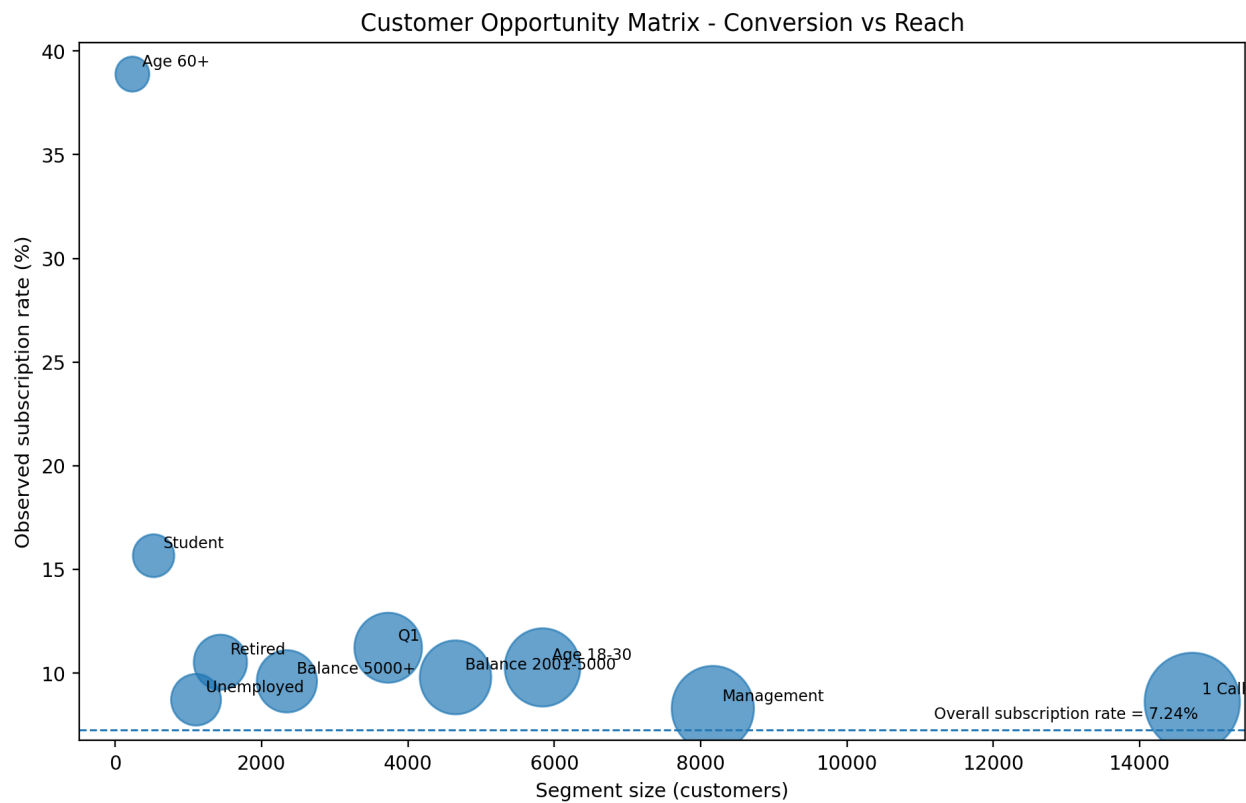
2. Requirement Coverage



Why this figure matters: Separates fully satisfied requirements from the one documentation gap: average 5-fold CV accuracy was not retained in the saved outputs.

Source: 01_Bank_Marketing_EDA.ipynb, 02_Random_Forest_Modeling.ipynb, 03_XGBoost_Modeling.ipynb, 04_Model_Comparison_and_Business_Analysis.ipynb.

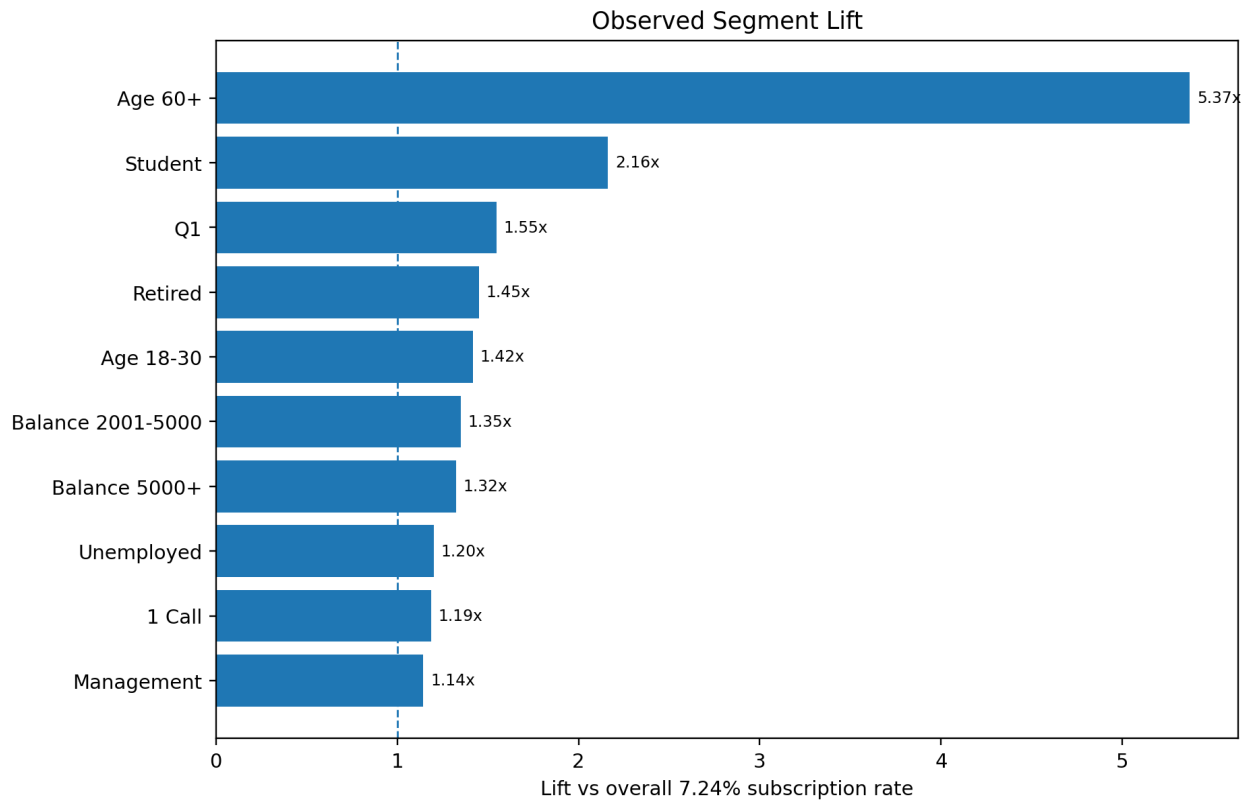
3. Customer Opportunity Matrix



Why this figure matters: Combines observed conversion rate with segment size. This is more decision-useful than selecting the highest conversion percentage alone.

Source: 01_Bank_Marketing_EDA.ipynb - Age, Job, Balance and Quarter analyses.

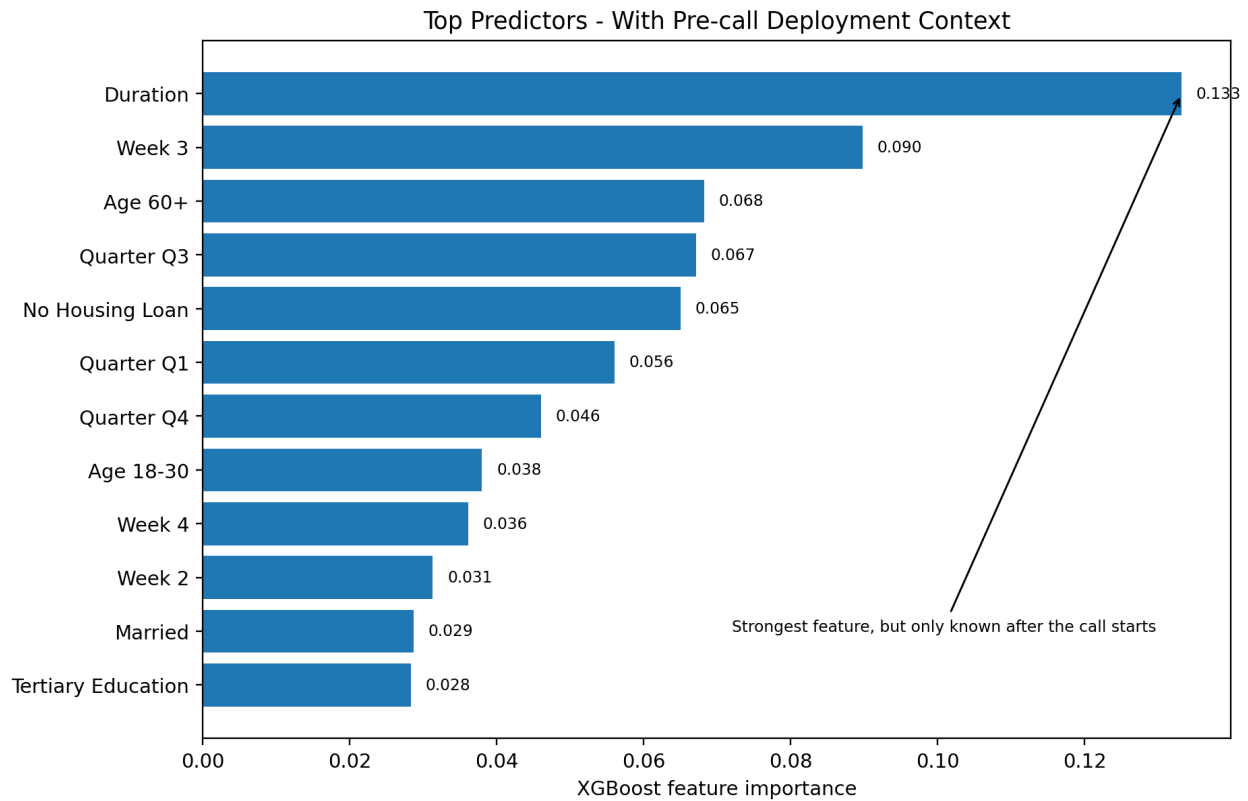
4. Segment Lift vs Overall Campaign



Why this figure matters: Shows how strongly selected segments over-index relative to the 7.24% overall subscription rate.

Source: 01_Bank_Marketing_EDA.ipynb - Target Distribution and segment analyses.

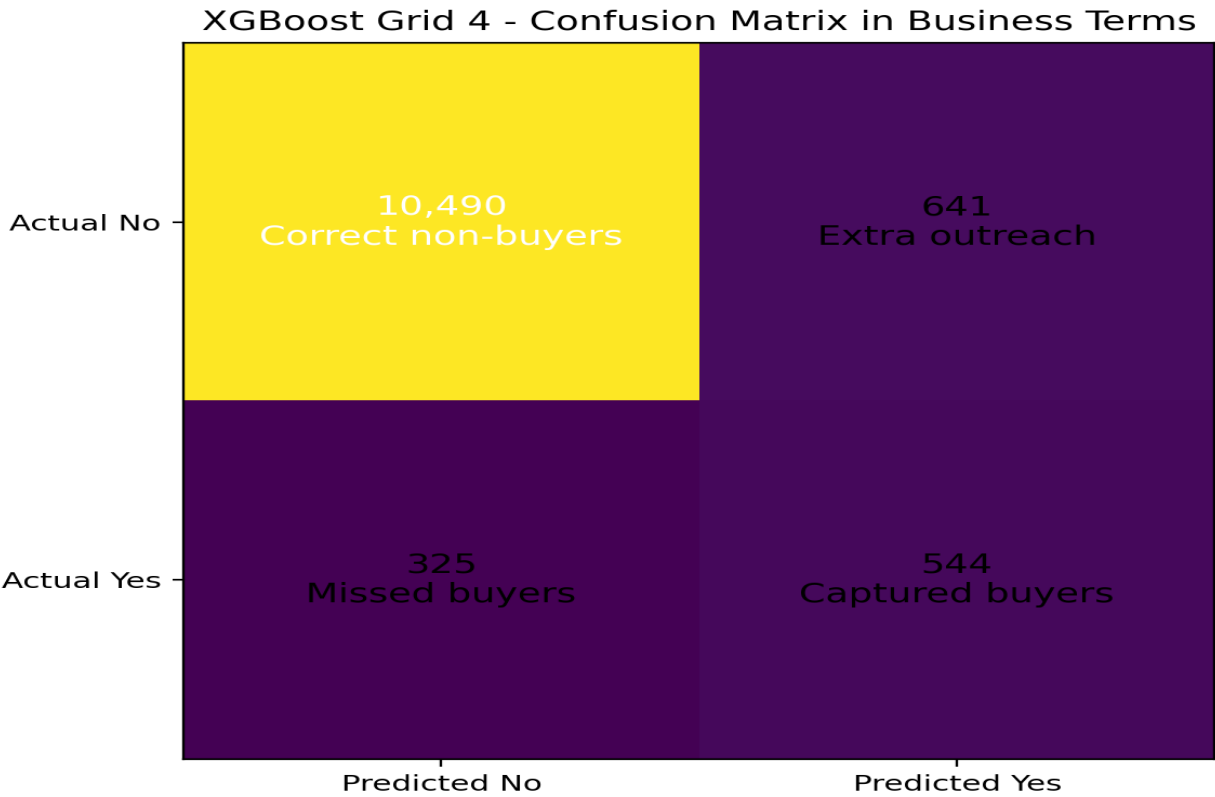
5. Predictors and Deployment Context



Why this figure matters: Shows the strongest XGBoost predictors while highlighting the key operational limitation: Duration is unavailable before the call.

Source: 03_XGBoost_Modeling.ipynb - Top 20 Feature Importance.

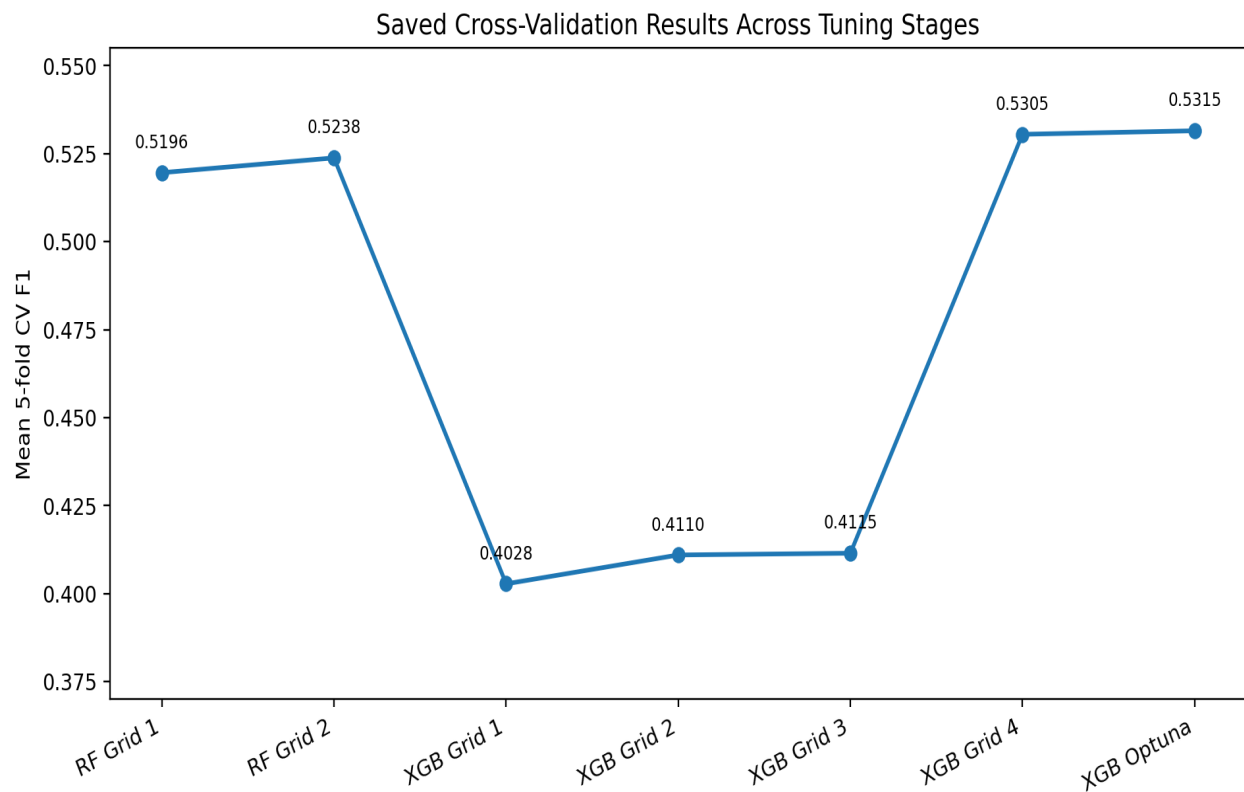
6. Confusion Matrix in Business Language



Why this figure matters: Translates classification outcomes into captured buyers, missed buyers, extra outreach and correct non-buyers.

Source: 04_Model_Comparison_and_Business_Analysis.ipynb - XGBoost Grid 4 Confusion Matrix.

7. Cross-Validation Tuning Progression



Why this figure matters: Shows the saved mean 5-fold CV F1 results and the small incremental gain from Grid 4 to Optuna.

Source: 02_Random_Forest_Modeling.ipynb and 03_XGBoost_Modeling.ipynb - saved GridSearch/Optuna outputs.